

CHAPTER FIVE

APPLICATION OF DERIVATIVES

At the end of this chapter students will be able to:

- ❖ Calculate Rate of change of different quantities
- ❖ Deal with Rolle's theorem and its application
- ❖ Mean value theorem and its application
- ❖ First derivative test and its application
- ❖ Second derivative test and its application
- ❖ Curve sketching
- ❖ L' Hoptitals Rule

5.1 Rate of Change

Suppose we have two variables x and y (in most problems the letters will be different, but for now let's use x and y) which are both changing with time. A “related rates” problem is a problem in which we know one of the rates of change at a given instant

say, $x' = \frac{dx}{dt}$ and we want to find the other rate $y' = \frac{dy}{dt}$ at that instant.

Here are the steps in doing a related rates problem:

1. Decide what the two variables are.
2. Find an equation relating them.
3. Take $\frac{d}{dt}$ of both sides.
4. Plug in all known values at the instant in question.
5. Solve for the unknown rate.

1. The radius of a circle is increasing at the rate of $5c \text{ m/s}$.

a) what is the rate of increase of its area when the radius 4cm?

b) What is the rate of increase of its circumference?

Solution: a) Known rate is $\frac{dr}{dt} = 5c \text{ m/s}$.

Unknown rate $\frac{dA}{dt}$.

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

But, $\frac{dr}{dt} = 5c \text{ m/s}$. Hence, $\frac{dA}{dt} = 2\pi r(5c \text{ m/s}) = 10\pi r c \text{ m/s}$.

$$\frac{dA}{dt} \big|_{r=4cm} = 10(4cm)c \text{ m/s} = 40c \text{ m}^2/\text{s}$$

$$\text{b) } C = 2\pi r \Rightarrow \frac{dC}{dt} = 2\pi \frac{dr}{dt}$$

$$\text{But, } \frac{dr}{dt} = 5c \text{ m/s. } \frac{dC}{dt} = 2\pi(5c \text{ m/s}) = 10\pi c \text{ m/s}$$

2. A ladder of 5m long is leaning against a wall. The bottom of the ladder is pulled along the ground away from the wall at the rate 20cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4m away from the wall?

Solution: Let 'y' be the height of the wall at which the ladder touches

Let the foot of the ladder be 'x' away from the wall.

By Pythagoras' theorem, we have

$$x^2 + y^2 = 25 \Rightarrow y = \sqrt{25 - x^2} \text{ (height is positive)}$$

$$\frac{dy}{dt} = \frac{-x}{\sqrt{25 - x^2}} \frac{dx}{dt}$$

$$\frac{dy}{dx} = \frac{-2x}{\sqrt{25 - x^2}} \text{ cm/s} \Rightarrow \frac{dy}{dt} \big|_{x=4\text{m}} = \frac{-8}{3} \text{ cm/s}$$

The negative rate of change indicates decreasing rate.

Exercise

1. Air is being pumped into spherical balloon at a rate of $20 \text{ cm}^3/\text{s}$. How fast is the radius of the balloon increasing when the radius is 6cm. Answer. $\frac{dr}{dt} = \frac{5}{36} \text{ cm/s}$.

2. A water tank has a shape inverted circular cone with base radius 2m and height 4m. If water is being pumped into the tank at the rate $2 \text{ m}^3/\text{s}$, then find the rate at which water level is rising when the water is 3m deep. Ans. $\frac{dh}{dt} = \frac{8}{9\pi}$.

5.2 Mean Value Theorem and its application

Before dealing with mean value theorem it preferable to see Rolle's theorem as follows.

Let f be a function satisfying the following three condition

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)
3. $f(a) = f(b)$, then there exist a number $c \in (a, b)$ such that $f'(c) = 0$.

Example: Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root.

Solution: Let $f(x) = x^3 + x - 1$

$$f(0) = -1 < 0$$

$$f(1) = 1 > 0$$

Since f is a polynomial, it is continuous, so the Intermediate value theorem states that there is a number c between 0 and 1 such that $f(c) = 0$.

Thus the given equation has a root.

- To show that the equation has no other real root, we use Rolle's Theorem and argue by contradiction.

Suppose that it had two roots a and b . Then $f(a) = 0 = f(b)$ and, since f is a polynomial, it is differentiable on (a, b) and continuous on $[a, b]$. Thus, by Rolle's theorem, there is a number c between a and b such that $f'(c) = 0$.

But $f'(x) = 3x^2 + 1 > 0$ so can never be 0. This gives a contradiction. Therefore the equation can't have two real roots.

Exercise

1. Determine whether Rolle's theorem can be applied. If so, find c . If not, explain why? $f(x) = x^4 - 2x^2$, $[-2, 2]$
2. Show that $x^5 + 4x = 1$ has exactly one solution.
3. Let $f(x) = x^{\frac{2}{3}}$ and note that $f(-1) = f(1) = 1$. Show that there is no $c \in [-1, 1]$ for which $f'(c) = 0$. Do you think that it contradicts Rolle's theorem????

5.2. Mean Value Theorem and its application

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

Proof: We introduce an auxiliary function g that allows us to simplify the proof by using Rolle's theorem. The function g is defined by

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right], a \leq x \leq b$$

Now g is continuous on $[a, b]$ and differentiable on (a, b) , Since g is a simple combination of f , constant function and a linear function. Substituting in the above, we find that

$$g(a) = g(b) = 0$$

So that by Rolle's theorem there is a number c in (a, b) such that $g'(c) = 0$.

However, $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$ for $a < x < b$

And thus $0 = g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}$

Solving for $f'(c)$, we obtain $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Supportive Example

1. Let $f(x) = \frac{1}{3}x^3 + 2x$. Find a number c in $(0,3)$ that mean value theorem.

Solution: $f(x)$ is continuous on $[0,3]$ and differentiable on $(0,3)$. Hence, by MVT we seek a number c in $(0,3)$ such that

$$f'(c) = \frac{f(3) - f(0)}{3 - 0} = \frac{15 - 0}{3 - 0} = 5$$

$$f'(x) = x^2 + 2 \Rightarrow f'(c) = c^2 + 2 = 5$$

Therefore, $c^2 = 3$, and since c must be in $(0,3)$, we conclude that $c = \sqrt{3}$.

Exercise

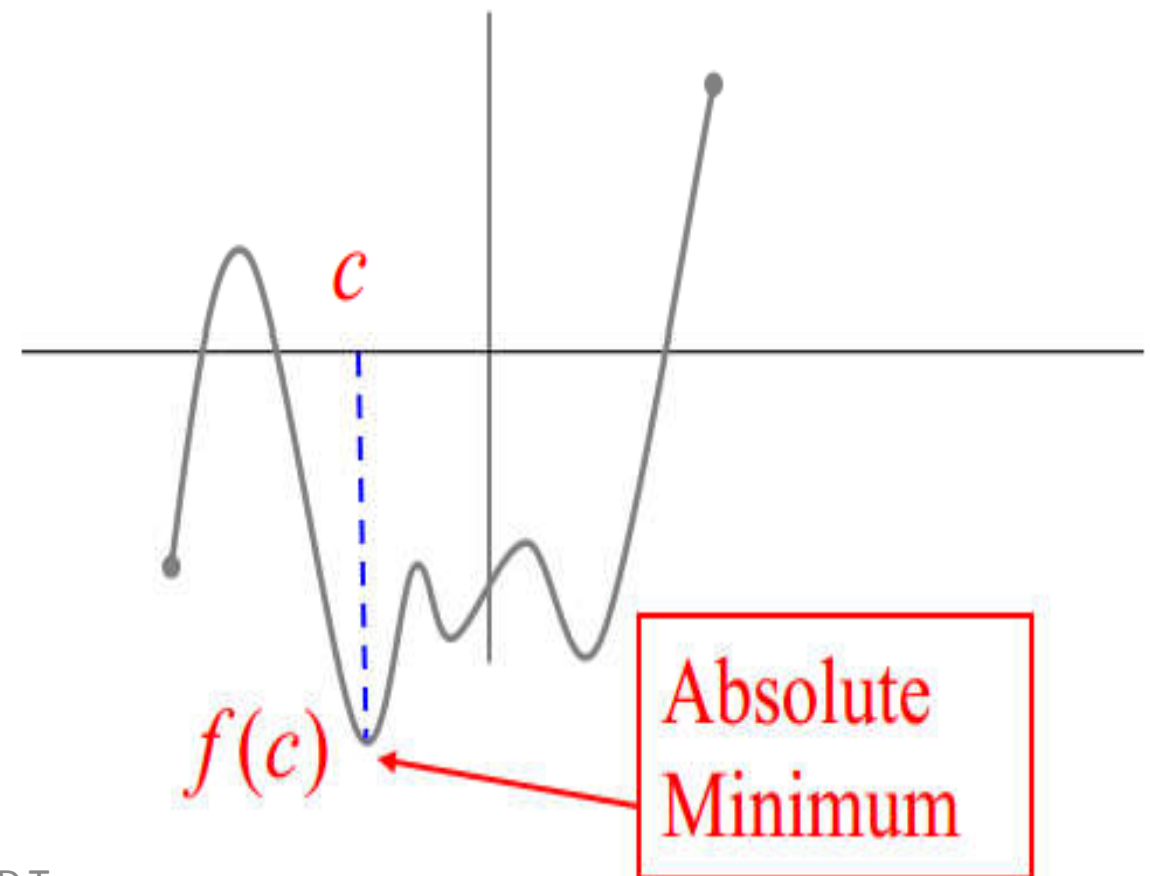
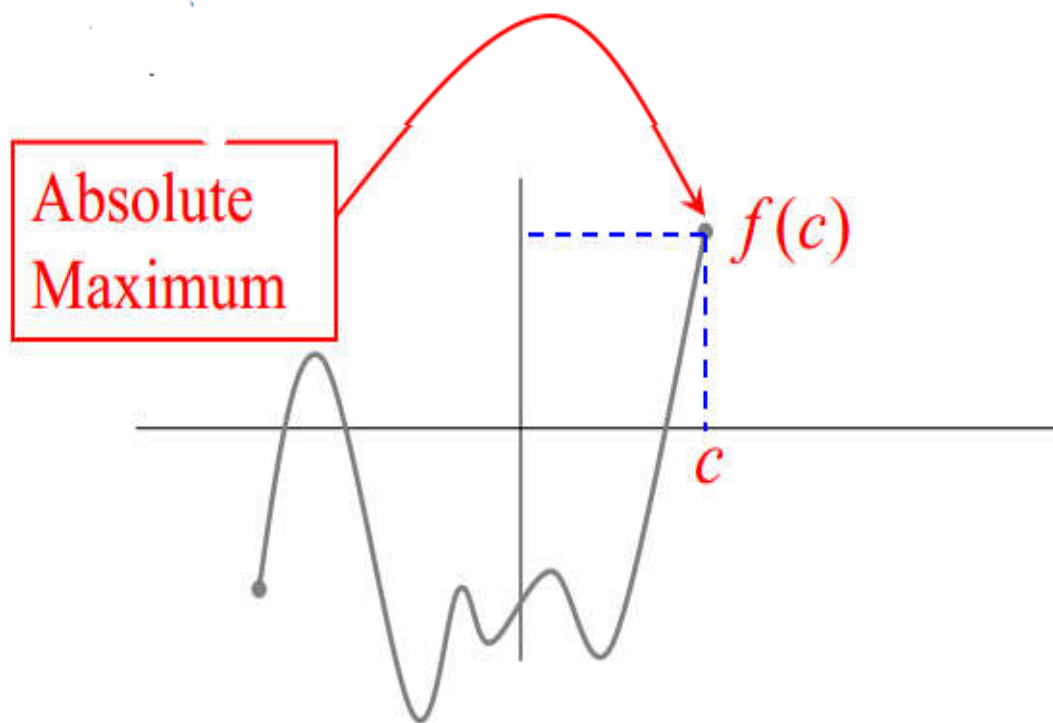
1. Find a number c that satisfy mean value theorem for $f(x) = x^3 + 12x^2 + 7x$ on the interval $[-4,4]$
2. Let $f(x) = Ax^2 + Bx + C$, where A, B , and C are constants with $A \neq 0$. Show that for any interval $[a, b]$, the number c guaranteed by Mean value theorem is the midpoint of $[a, b]$.

5.3.The First and Second Derivatives and Its Application

Absolute Extrema

A function f has an **absolute (global) maximum** at $x = c$ if $f(x) \leq f(c)$ for all x in the domain D of f . The number $f(c)$ is called the *absolute maximum* value of f in D .

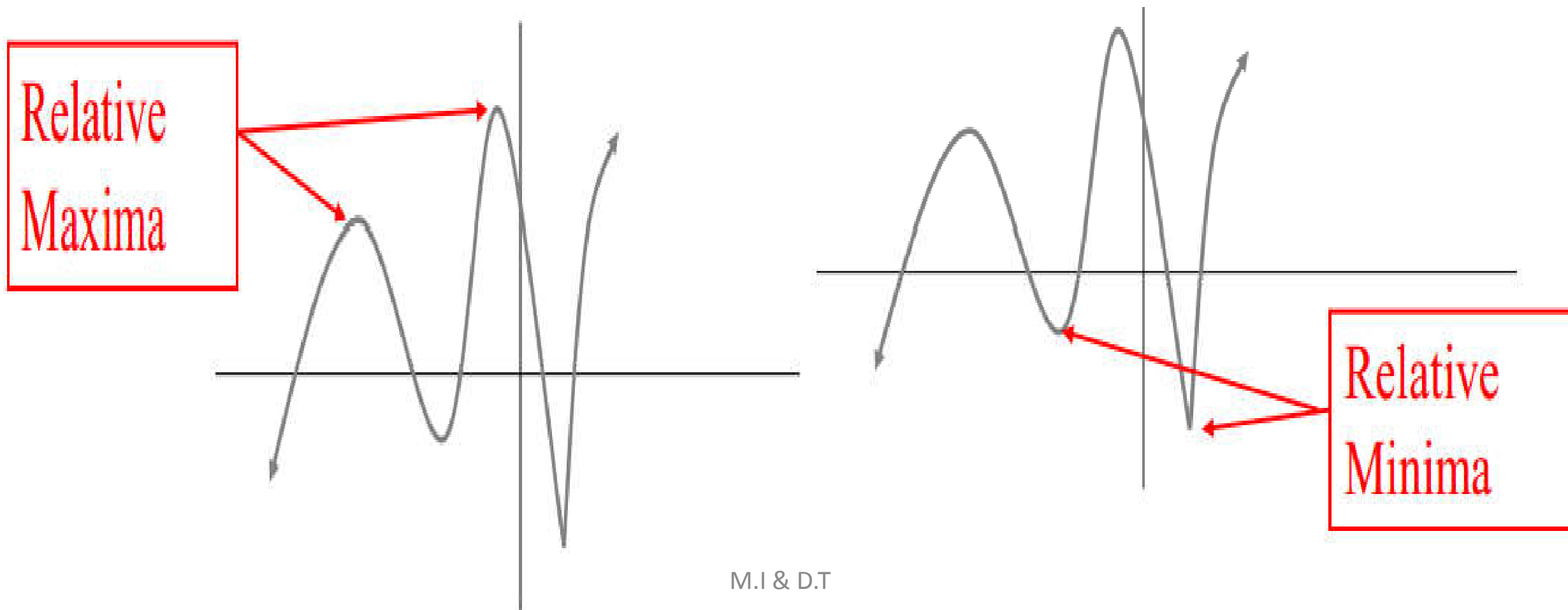
A function f has an **absolute (global) minimum** at $x = c$ if $f(c) \leq f(x)$ for all x in the domain D of f . The number $f(c)$ is called the *absolute minimum* value of f in D .



Relative Extrema

A function f has a **relative (local) maximum** at $x = c$ if there exists an open interval (a, b) containing c such that $f(x) \leq f(c)$ for all $a \leq x \leq b$.

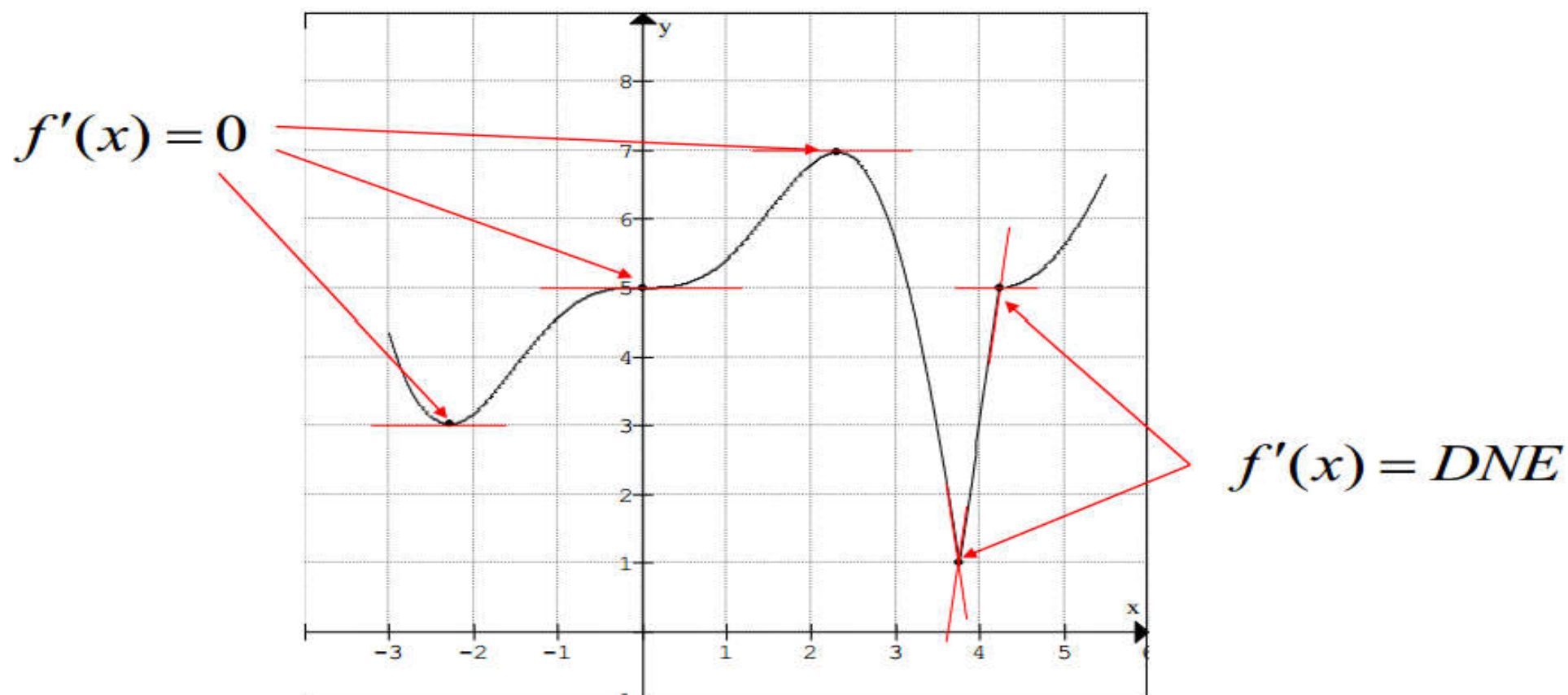
A function f has a **relative (local) minimum** at $x = c$ if there exists an open interval (a, b) containing c such that $f(c) \leq f(x)$ for all $a \leq x \leq b$.



Critical Points of f

A critical number of a function f is a number c in the domain of f such that

- a) $f'(c) = 0$ (stationary point)
- b) $f'(c)$ does not exist (DNE) (singular point)



The corresponding values of x are called
Critical Points of f

Example 1. Find all the critical numbers of $f(x) = \sqrt[3]{x^3 - 3x}$

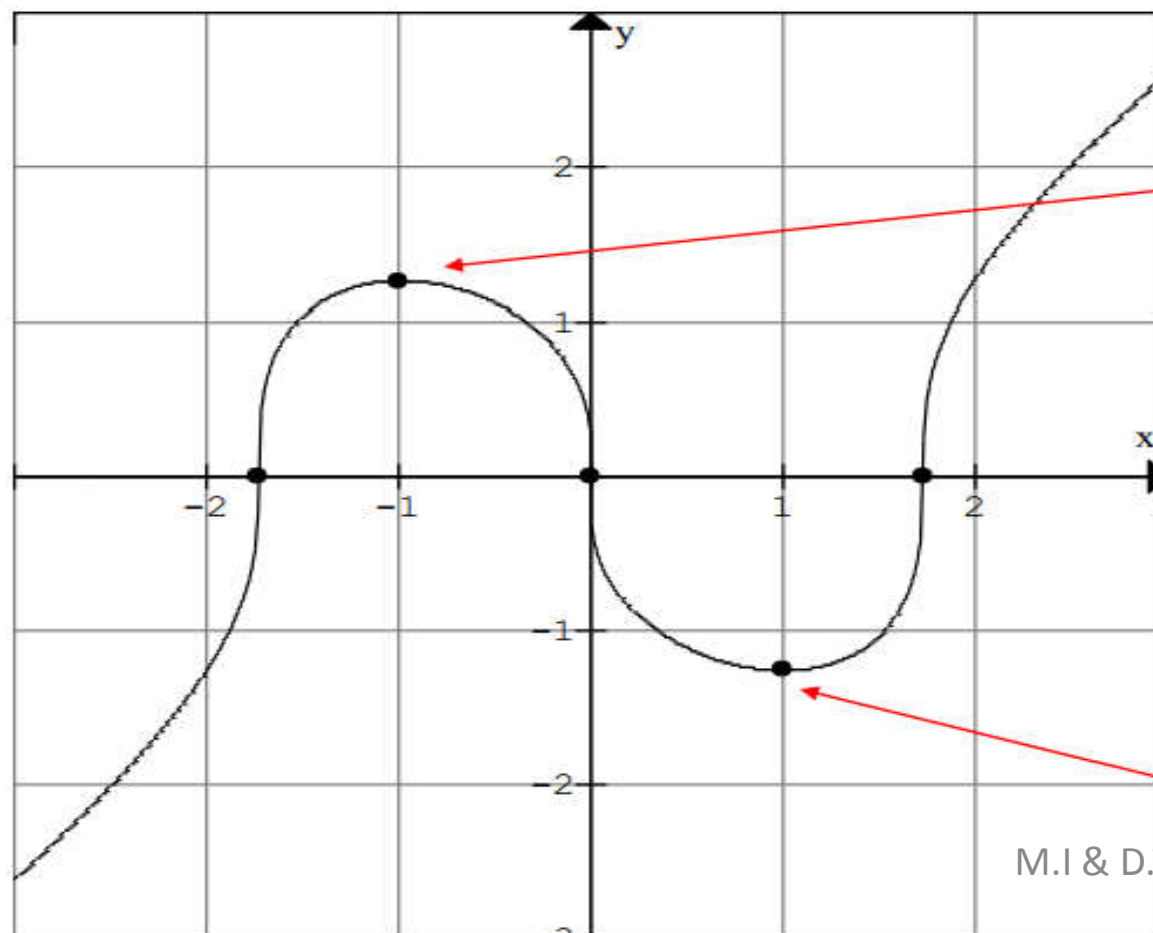
Solution:

$$f(x) = \sqrt[3]{x^3 - 3x}$$

$$\Rightarrow f'(x) = \frac{x^2 - 1}{\sqrt[3]{(x^3 - 3x)^2}}$$

Stationary point: $x = \pm 1$

Singular point: $x = 0, \pm\sqrt{3}$



Local max. $f(-1) = \sqrt[3]{2}$

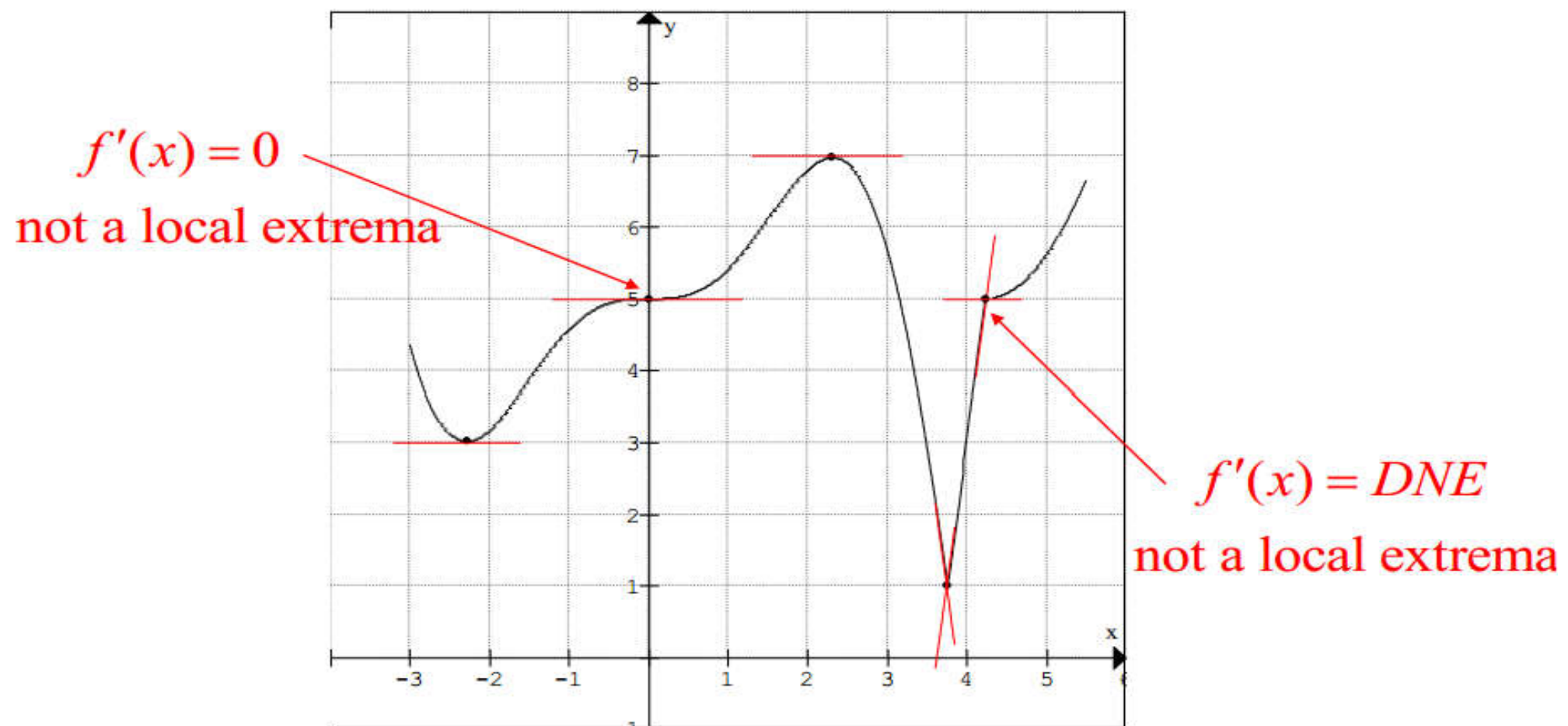
graph of $f(x) = \sqrt[3]{x^3 - 3x}$

Local min. $f(1) = -\sqrt[3]{2}$

Fermat's Theorem

If a function f has a local maximum or minimum at c , then c is a critical number of f

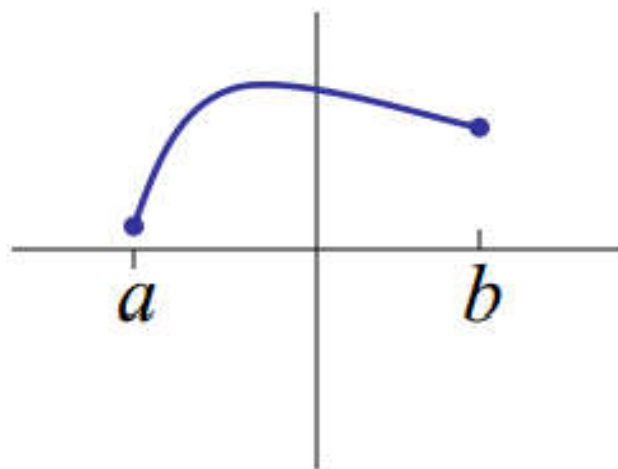
Notice that the theorem does not say that at every critical number the function has a local maximum or local minimum.



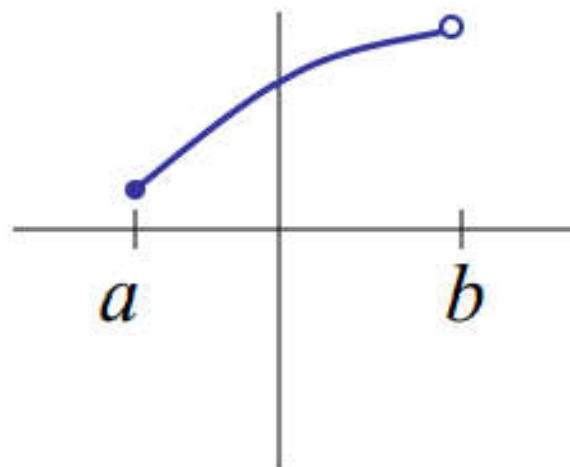
Two critical points of f that do not correspond to local extrema

Extreme Value Theorem

If a function f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum and absolute minimum on $[a, b]$. Each extremum occurs at a critical number or at an endpoint

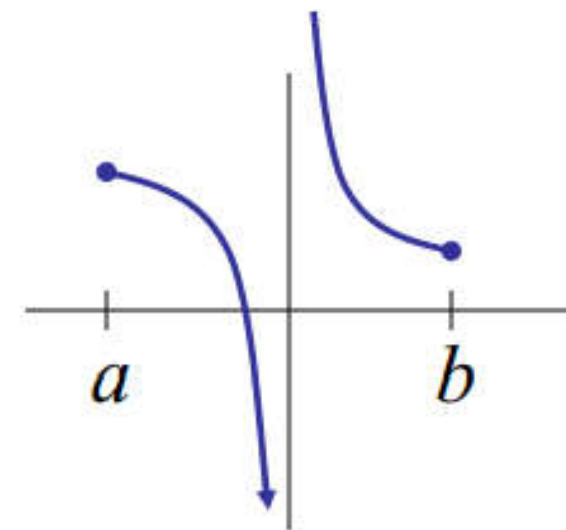


Attains max.
and min.



Attains min.
but no max.

Open Interval



No min. and
no max.

Not continuous

Finding absolute extrema on $[a, b]$

1. Find all critical numbers for $f(x)$ in (a, b) .
2. Evaluate $f(x)$ for all critical numbers in (a, b) .
3. Evaluate $f(x)$ for the endpoints a and b of the interval $[a, b]$.
4. The largest value found in steps 2 and 3 is the **absolute maximum** for f on the interval $[a, b]$, and the smallest value found is the **absolute minimum** for f on $[a, b]$.

Example 2. Find the absolute extrema of $f(x) = x^3 - 3x^2$ on $\left[-\frac{1}{2}, 3\right]$

Solution:

Critical values of f **inside the interval** $(-1/2, 3)$ are $x = 0, 2$

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

Evaluate

$$f(0) = 0$$

Absolute Max.

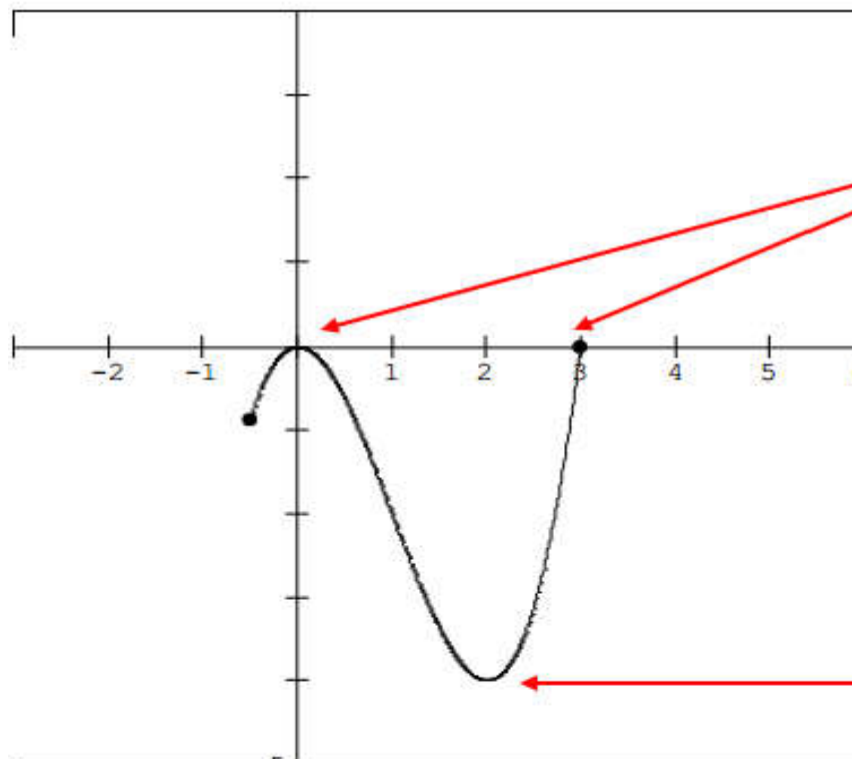
$$f(2) = -4$$

Absolute Min.

$$f\left(-\frac{1}{2}\right) = -\frac{7}{8}$$

$$f(3) = 0$$

Absolute Max.



Absolute Max.

Absolute Min.

Activity 1.

1. Find all the critical numbers of the following function

a) $f(x) = \frac{x+1}{x^2+1}$ **b)** $f(x) = x^2 e^{2x}$ **c)** $f(x) = \begin{cases} x^2 + 2x - 1 & \text{if } x < 1 \\ x^2 - 4x + 3 & \text{if } x \geq 0 \end{cases}$

d) $f(x) = \sqrt{1-x^2}$ **e)** $f(x) = x^{2/3}(x-1)^2$ **f)** $f(x) = (x+5)\sqrt[3]{x-4}$

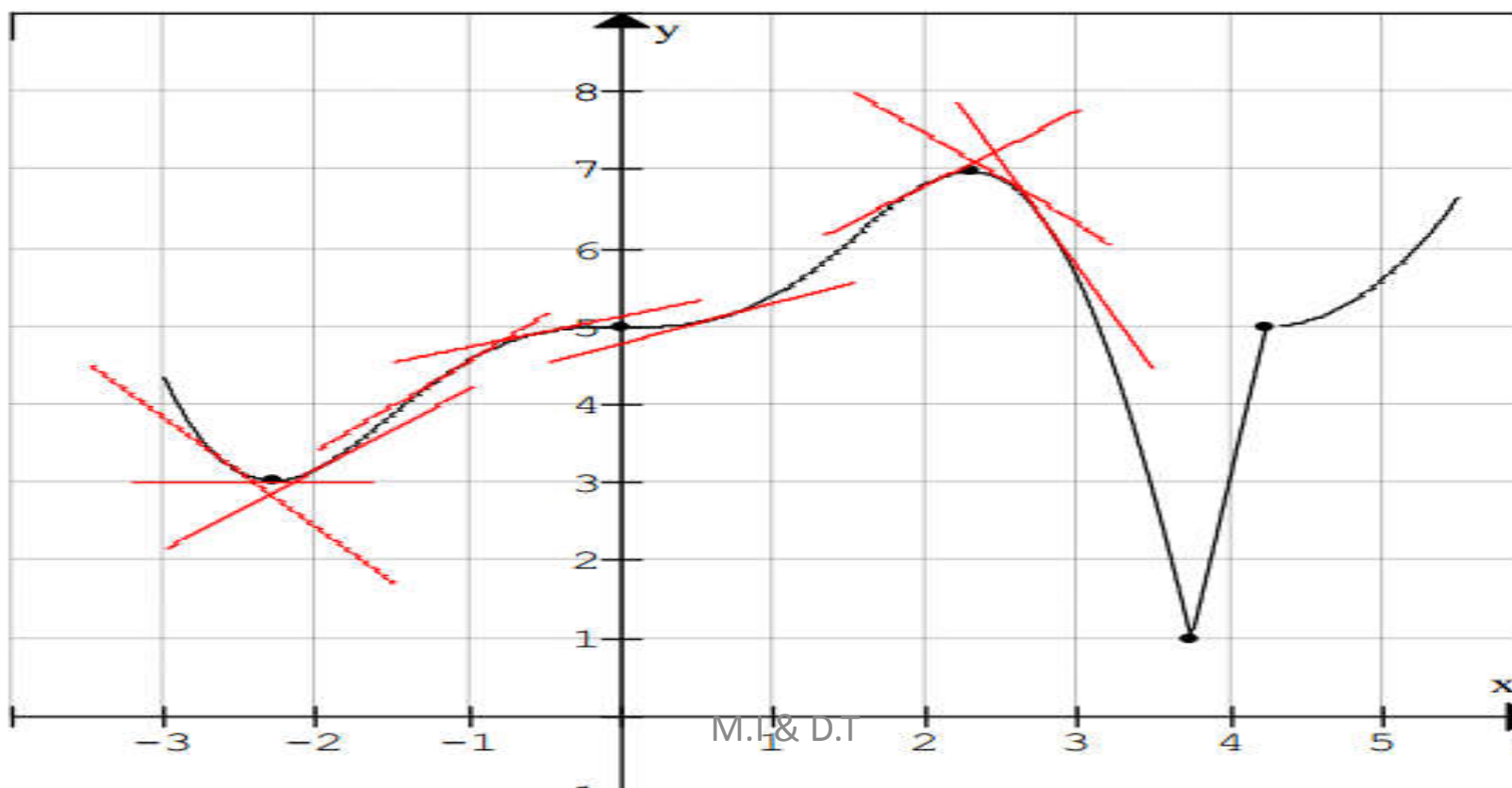
2. Find the Absolute extrema of the following function on the indicated interval I

a) $f(x) = 2x^3 - 3x^2 - 12x + 5, \quad I = [-2, 4]$ **b)** $f(x) = \frac{3x^2}{x-3}, \quad I = [-2, 2]$

c) $f(x) = 4x^{5/4} - 8x^{1/4}, \quad I = [0, 4]$ **d)** $f(x) = 2\sin x - \cos x, \quad I = [0, 2\pi]$

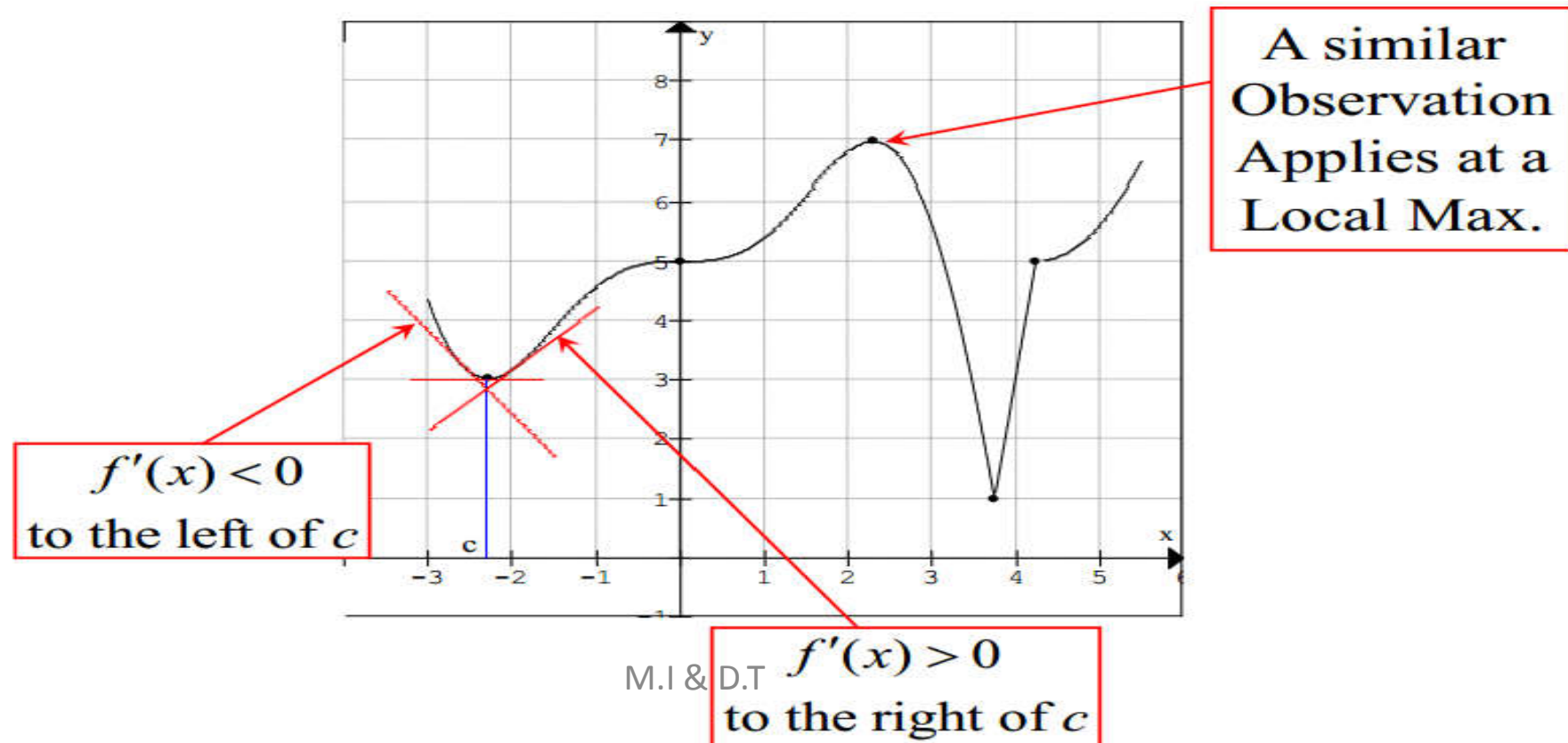
Increasing, Decreasing and Constant

- If $f'(x) > 0$ for each value of x in an interval (a, b) , then f is **increasing** on (a, b) .
- If $f'(x) < 0$ for each value of x in an interval (a, b) , then f is **decreasing** on (a, b) .
- If $f'(x) = 0$ for each value of x in an interval (a, b) , then f is **constant** on (a, b) .



The first Derivative test: Suppose that c is a critical number of a continuous function f

- If f' changes from positive to negative at c , then f has a **local maximum** at c .
- If f' changes from negative to positive at c , then f has a **local minimum** at c .
- If f' does not change sign at c (for example, if f' is positive on both side or negative on both side), then f has **no local maximum or local minimum** at c .



Determine the sign of the derivative of f to the left and right of the critical point

left	right	conclusion
+	−	$f(c)$ is a relative maximum
−	+	$f(c)$ is a relative minimum
No change		No relative extremum

Example 3. Find all the relative extrema of $f(x) = x^3 - 6x^2 + 1$

Solution: $f'(x) = 3x^2 - 12x$

$$f'(x) = 3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

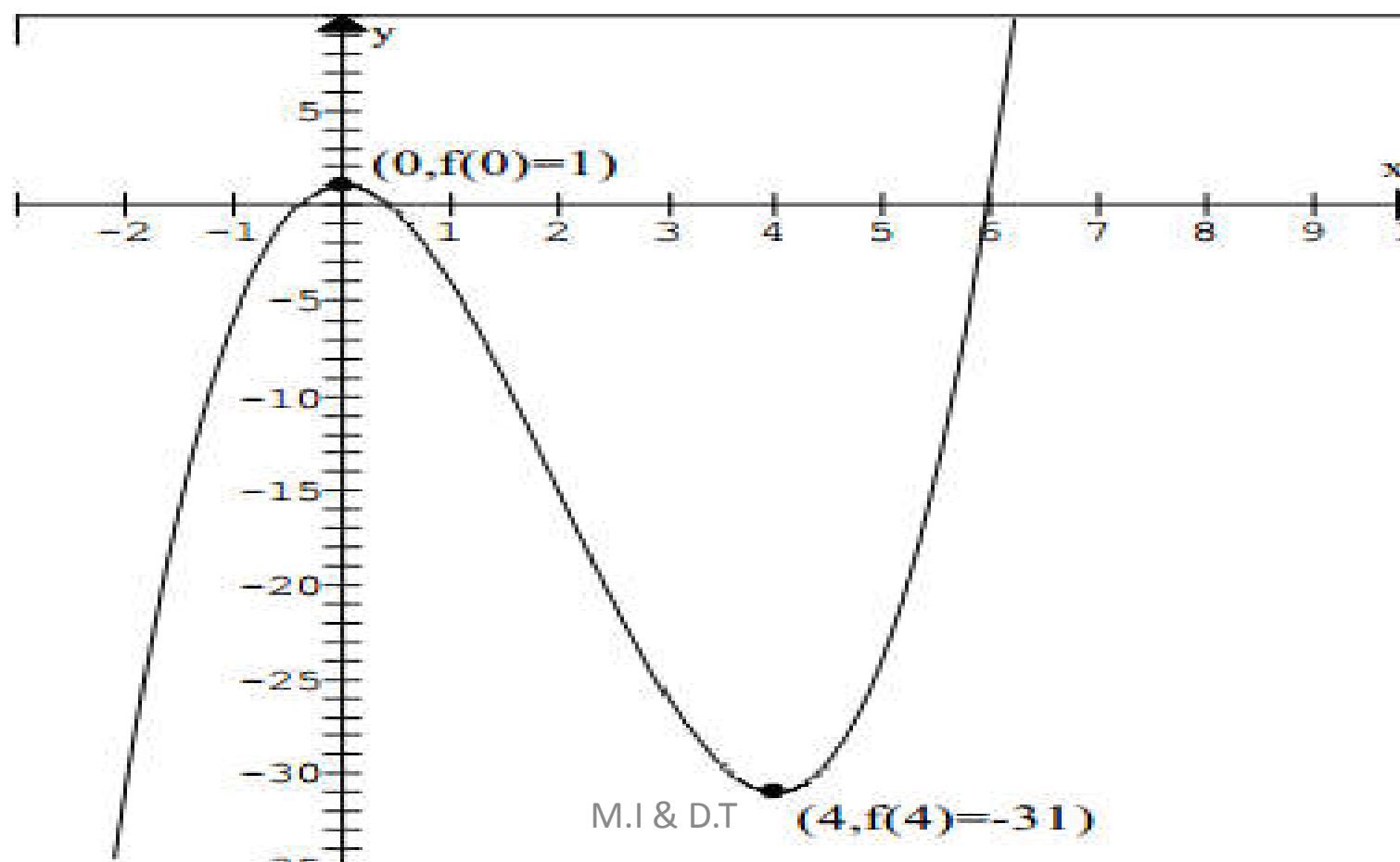
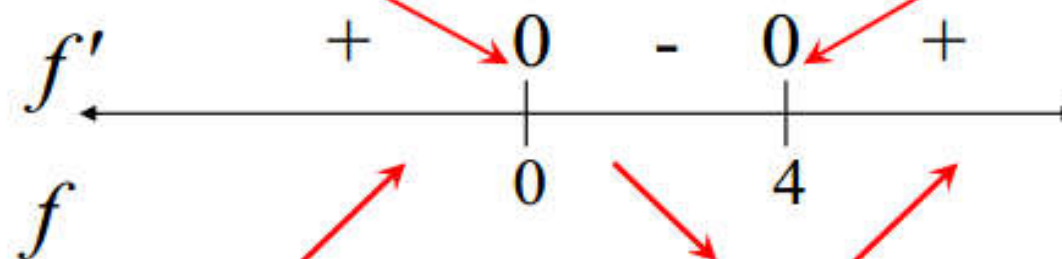
$$x = 0, 4$$

Stationary point: $x = 0, 4$

Singular point: None

Relative max.
 $f(0) = 1$

Relative min.
 $f(4) = -31$



Example 4. Find all the relative extrema of $f(x) = \sqrt[3]{x^3 - 3x}$

Solution:

$$f(x) = \sqrt[3]{x^3 - 3x} \Rightarrow f'(x) = \frac{x^2 - 1}{\sqrt[3]{(x^3 - 3x)^2}}$$

Stationary point: $x = \pm 1$

Singular point: $x = 0, \pm\sqrt{3}$

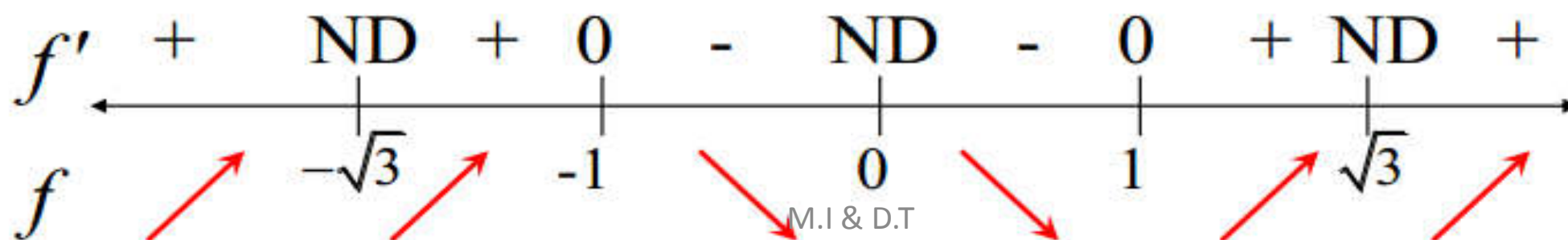
Relative max.

$$f(-1) = \sqrt[3]{2}$$

Graph on page 17

Relative min.

$$f(1) = -\sqrt[3]{2}$$



Optimization Problems

A Procedure for Solving Applied Maximum and Minimum Problems

Step 1. Draw an appropriate figure and label the quantities relevant to the problem.

Step 2. Find a formula for the quantity to be maximized or minimized.

Step 3. Using the conditions stated in the problem to eliminate variables, express the quantity to be maximized or minimized as a function of one variable.

Step 4. Find the interval of possible values for this variable from the physical restrictions in the problem.

Step 5. If applicable, use the techniques of the preceding section to obtain the maximum or minimum.

Example 5. A garden is to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 100 running feet of chicken wire is available for the fence?

Solution:

Let x = length of the rectangle (ft)

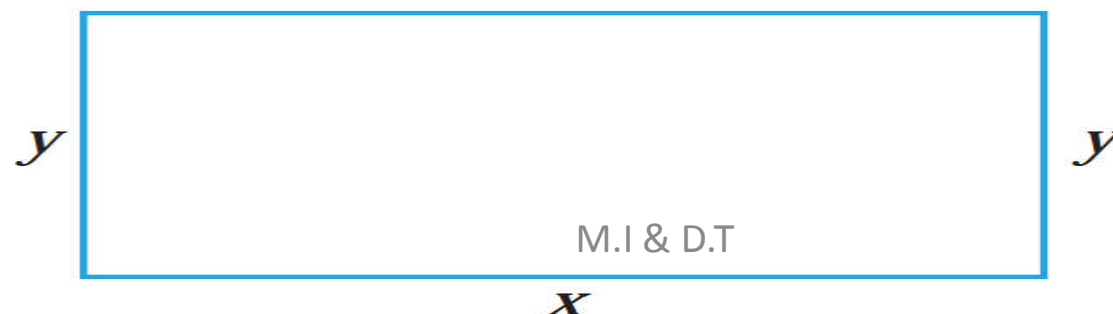
y = width of the rectangle (ft)

A = area of the rectangle (ft²)

Then $A = xy$ [1]

Since the perimeter of the rectangle is 100 ft, the variables x and y are related by the equation

$$2x + 2y = 100 \quad \text{or} \quad y = 50 - x \quad [2]$$



Substituting [2] in [1] yields

$$A = x(50 - x) = 50x - x^2 \quad [3]$$

Because x represents a length, it cannot be negative, and because the two sides of length x cannot have a combined length exceeding the total perimeter of 100 ft, the variable x must satisfy

$$0 \leq x \leq 50 \quad [4]$$

Thus, we have reduced the problem to that of finding the value (or values) of x in $[0, 50]$, for which A is maximum.

Since A is a polynomial in x , it is continuous on $[0, 50]$, and so the maximum must occur at an endpoint of this interval or at a critical point.

From [3] we obtain $dA/dx = 50 - 2x$

Setting $dA/dx = 0$ we obtain $50 - 2x = 0$ or $x = 25$.

Thus, the maximum occurs at one of the values $x = 0$, $x = 25$, $x = 50$

Substituting these values in [3] yields the below table

x	0	25	50
A	0	625	0

which tells us that the maximum area of 625 ft² occurs at $x = 25$. From [2] the corresponding value of y is 25, so the rectangle of perimeter 100 ft with greatest area is a square with sides of length 25 ft

Activity 2.

1. Find local maximum and minimum value of the following function

a) $f(x) = x^{2/3}(x-1)^2$ **b)** $f(x) = \frac{(a+x)}{\sqrt{ax}}, \quad x > 0 \text{ and } a \neq 0$

2. If $f(x) = ax^3 + bx^2 + cx + d$, determine the values of a, b, c and d, such that f has local maximum 2 at $x = -1$ and a local maximum -1 at $x = 1$

3. If $f(x) = ax^4 + bx^3 + cx^2 + dx + e$, determine the values of a, b, c d and e, such that f has local maximum 2 at $x = 0$, and a local maximum -14 at $x = -2$ and $x = 2$

4. Find the radius and height of the right circular cylinder of largest volume that can be inscribed in a right circular cone with radius 6 inches and height 10 inches

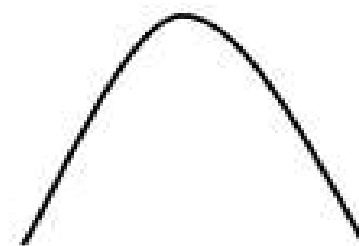
Concavity

Let f be a differentiable function on (a, b) .

1. f is **concave upward** on (a, b) if f' is increasing on (a, b) . That is $f''(x) > 0$ for each value of x in (a, b)
2. f is **concave downward** on (a, b) if f' is decreasing on (a, b) . That is $f''(x) < 0$ for each value of x in (a, b)



concave upward



concave downward

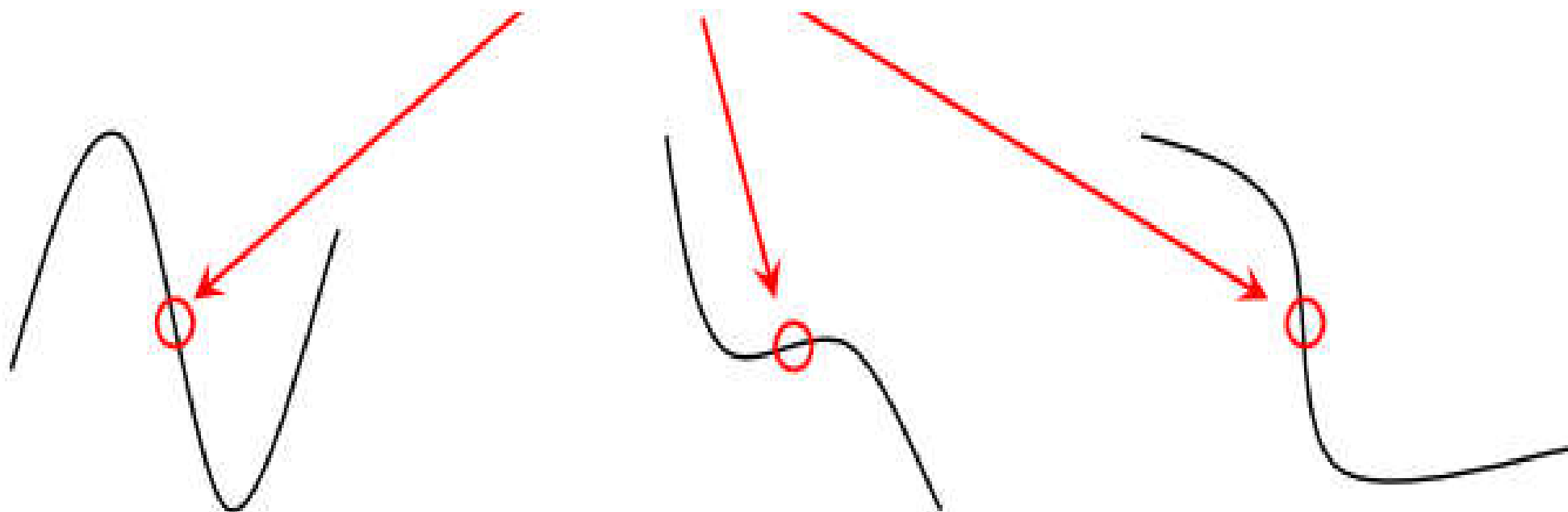
Example 6. determine open interval on which the graph of $f(x)$ is concave upward and downward

a) $f(x) = x^3 + x^2 - 5x - 5$

b) $f(x) = \frac{x^2 + 1}{x^2 - 4}$

Inflection Point

A point on the graph of f at which f is continuous and concavity changes is called an **inflection point**.



To search for inflection points, find any point, c in the domain where $f''(x) = 0$ or $f''(x)$ is undefined. If f'' changes sign from the left to the right of c , then $(c, f(c))$ is an inflection point of f .

Example 7. Find all inflection points of $f(x) = x^3 - 6x^2 + 2$

Solution : let $f(x) = x^3 - 6x^2 + 2$

$$f'(x) = 3x^2 - 12x$$

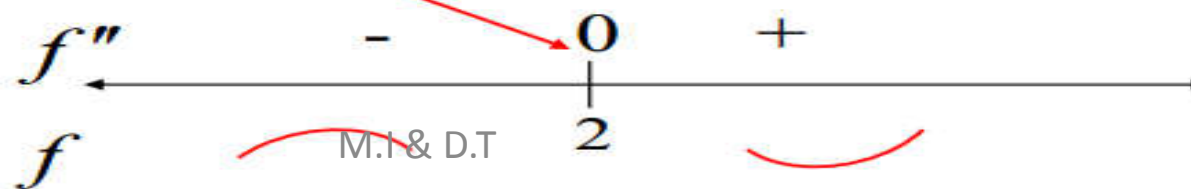
$$f''(x) = 6x - 12$$

$$6x - 12 = 0$$

$$x = 2$$

Thus , since the polynomial function is defined at every point, the only Possible inflection point is $x = 2$.

Inflection
point at $x = 2$



5.4 Curve Sketching

Properties of Graphs

- In many problems, the properties of interest in the graph of a function are:
 - Domain
 - x and y -intercepts
 - Local maximum or minimum
 - Intervals of increase and decrease
 - Symmetry
 - Asymptotes
 - Concavity
 - Inflection points
- Recall that a rational function is a function of the form $f(x) = P(x)/Q(x)$ in which $P(x)$ and $Q(x)$ are polynomials. Graphs of rational functions are more complicated than those of polynomials because of the possibility of asymptotes and discontinuities.

Graphing a Rational Function

$f(x) = P(x)/Q(x)$, if $P(x)$ and $Q(x)$ have no Common Factors

Step 1. (Domain): This is the set of values of x for which $f(x)$ is defined.

Step 2. (symmetries). Determine whether there is symmetry about the y -axis or the origin.

- **Symmetry-Even function**

If $f(-x) = f(x)$ for all x in D , that is, the equation of the curve is unchanged when x is replaced by $-x$, then f is an even function and the curve is symmetric about the y -axis.

- **Symmetry-Odd function**

If $f(-x) = -f(x)$ for all x in D , then f is an odd function and the curve is symmetric about the origin.

Note that: If neither $f(-x) = f(x)$ nor $f(-x) = -f(x)$ for all x in D , then a function f is neither even nor odd.

Step 3. (x- and y-intercepts). Find the x - and y -intercepts.

- The y -intercept is $f(0)$ and this tells us where the curve intersects the y -axis and to find the x -intercepts, we set $y = 0$ and solve for x .

Step 4. (asymptotes).

- Find the values of x for which $Q(x) = 0$. The graph has a **vertical asymptote** at each such value.
- If either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$ then the line $y = L$ is a **horizontal asymptote** of the curve $y = f(x)$
- **Oblique asymptotes** are discussed at the end of this section

Step 5. (derivatives). Find $f'(x)$ and $f''(x)$.

- **Interval of Increasing or Decreasing**
 - Compute $f'(x)$ and find the intervals on which:
 - $f'(x)$ is positive (f is increasing).
 - $f'(x)$ is negative (f is decreasing)
- **Local maximum and minimum**
 - Find the critical numbers of f (the numbers c where $f'(c) = 0$ or $f'(c)$ does not exist). Then, use the First Derivative Test.
 - If f' changes from **positive to negative** at a critical number c , then $f(c)$ is a local maximum.
 - If f' changes from **negative to positive** at c , then $f(c)$ is a local minimum.
- Although it is usually preferable to use the First Derivative Test, you can use the **Second Derivative Test** if $f'(c) = 0$ and $f''(c) \neq 0$. Then,
 - $f''(c) > 0$ implies that $f(c)$ is a local minimum.
 - $f''(c) < 0$ implies that $f(c)$ is a local maximum.
- **Concavity:** The curve is:
 - Concave upward where $f''(x) > 0$
 - Concave downward where $f''(x) < 0$

Step 6. (sketch the graph).

- Using the information above (step1-5), draw the graph. Sketch the asymptotes as dashed lines. Plot the intercepts, maximum and minimum points, and inflection points. Then, make the curve pass through these points, rising and falling according to step 5 (interval of increasing of decreasing), with concavity , and approaching the asymptotes.

Example 8: sketch the curve $y = \frac{2x^2}{x^2 - 1}$

Solution:

Step 1. The domain is: $\{x \mid x^2 - 1 \neq 0\} = \{x \mid x \neq \pm 1\} = (-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.

Step 2. Since $f(-x) = f(x)$, the function is even.

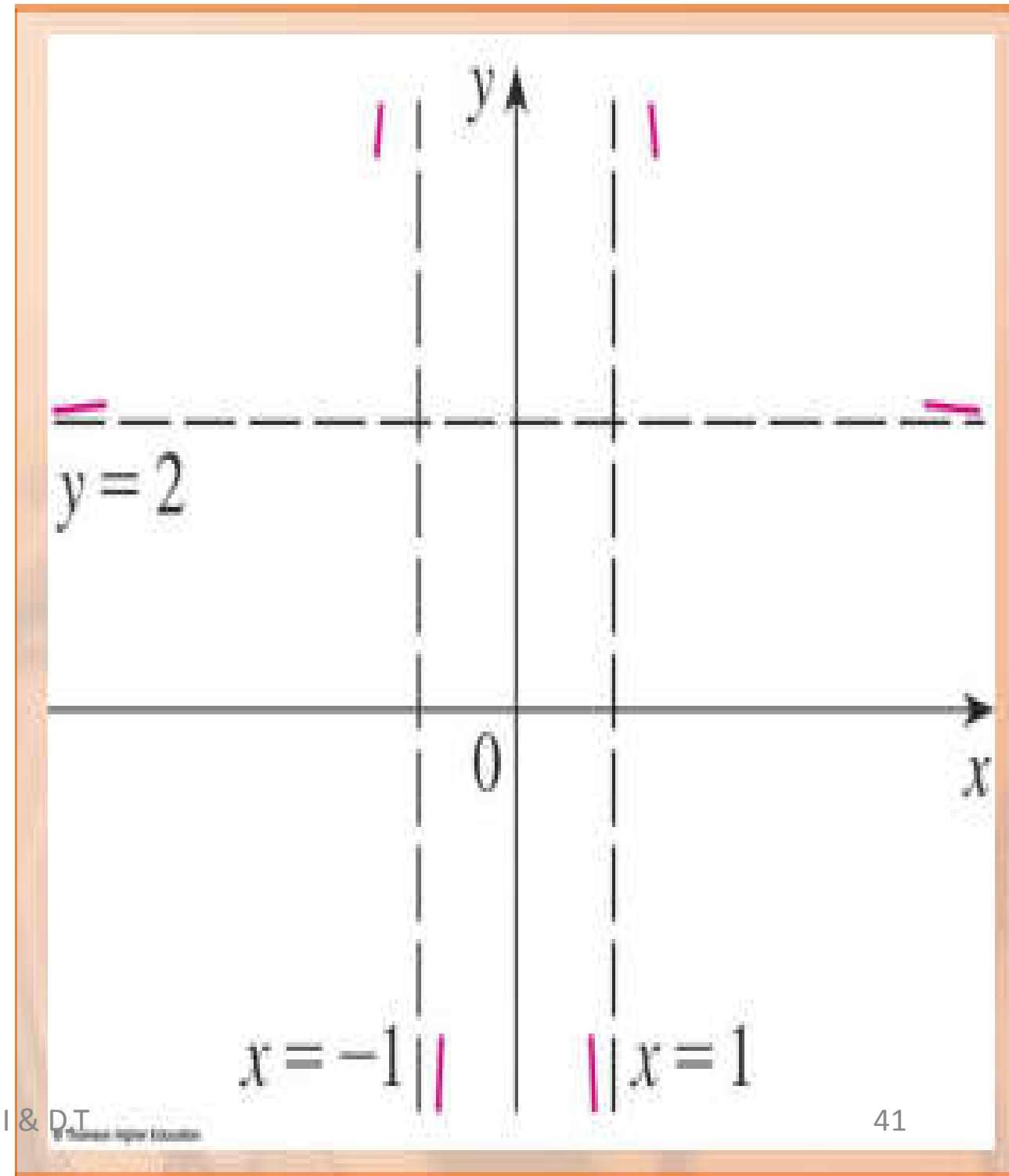
- The curve is symmetric about the y-axis

Step 3. The x- and y-intercepts are both 0.

Step 4. $\lim_{x \rightarrow \pm \infty} \frac{2x^2}{x^2 - 1} = \lim_{x \rightarrow \pm \infty} \frac{2}{1 - 1/x^2} = 2$ Therefore, the line $y = 2$ is a horizontal asymptote

Since the $Q(x) = 0$ when $x = \pm 1$. Thus, the lines $x = 1$ and $x = -1$ are vertical asymptotes.

This information about limits and asymptotes enables us to draw the preliminary sketch, showing the parts of the curve near the asymptotes



Step 5.
$$f'(x) = \frac{4x(x^2 - 1) - 2x^2 \cdot 2x}{(x^2 - 1)^2} = \frac{-4x}{(x^2 - 1)^2}$$

Since $f'(x) > 0$ when $x < 0$ ($x \neq 1$) and $f'(x) < 0$ when $x > 0$ ($x \neq 1$),

f is **Increasing** on $(-\infty, -1)$ and $(-1, 0)$ and **Decreasing** on $(0, 1)$ and $(1, \infty)$.

- It has no point of inflection since 1 and -1 are not in the domain of f .

The only critical number is $x = 0$. Since f' changes from positive to negative at 0, $f(0) = 0$ is a local maximum by the First Derivative Test

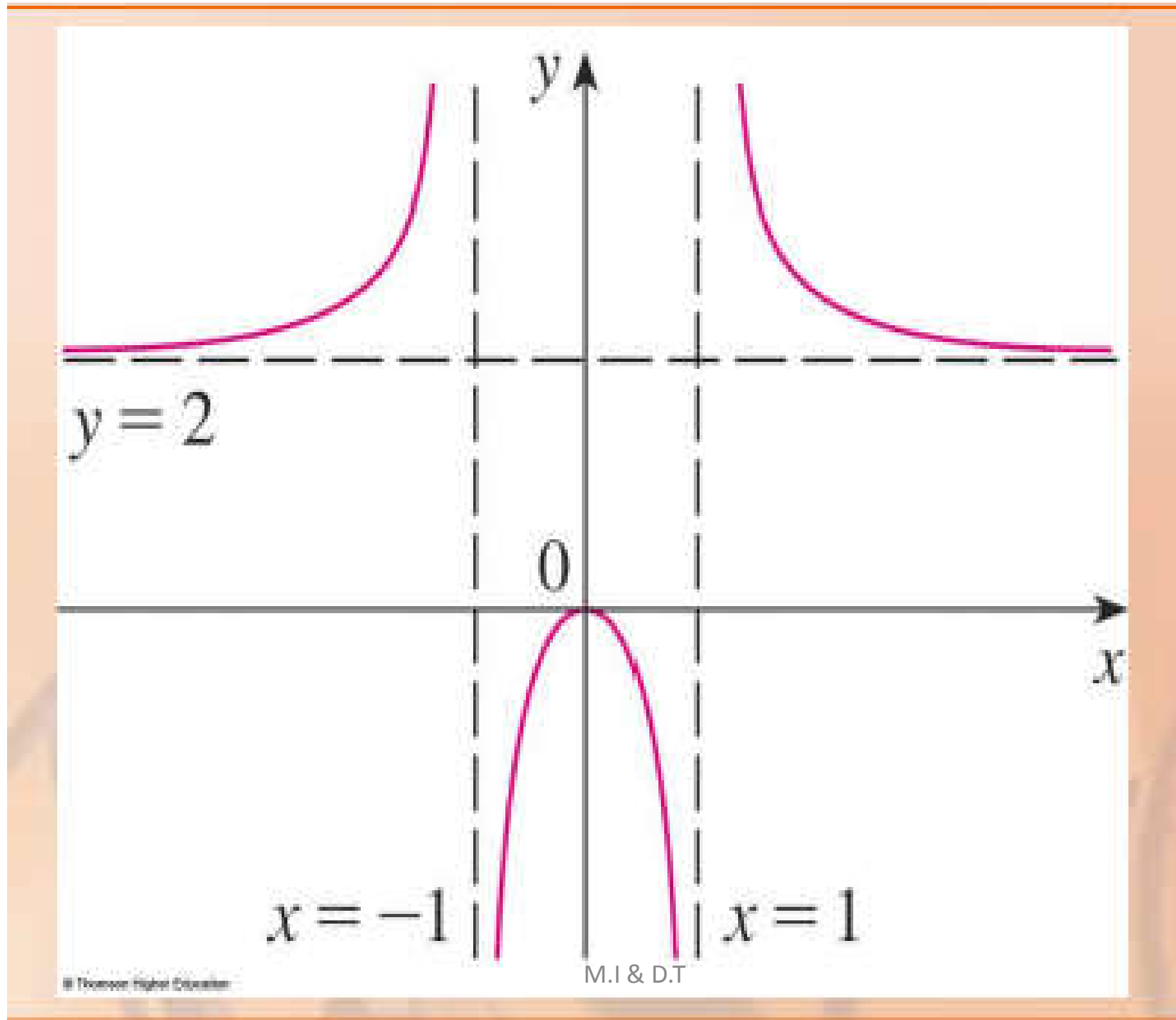
$$f''(x) = \frac{-4(x^2 - 1)^2 + 4x \cdot 2(x^2 - 1)2x}{(x^2 - 1)^4} = \frac{12x^2 + 4}{(x^2 - 1)^3}$$

Since $12x^2 + 4 > 0$ for all x , we have

$$f''(x) > 0 \Leftrightarrow x^2 - 1 > 0 \Leftrightarrow |x| > 1 \text{ and } f''(x) < 0 \Leftrightarrow |x| < 1$$

Thus, the curve is concave upward on the intervals $(-\infty, -1)$ and $(1, \infty)$ and concave downward on $(-1, 1)$

Step 6.

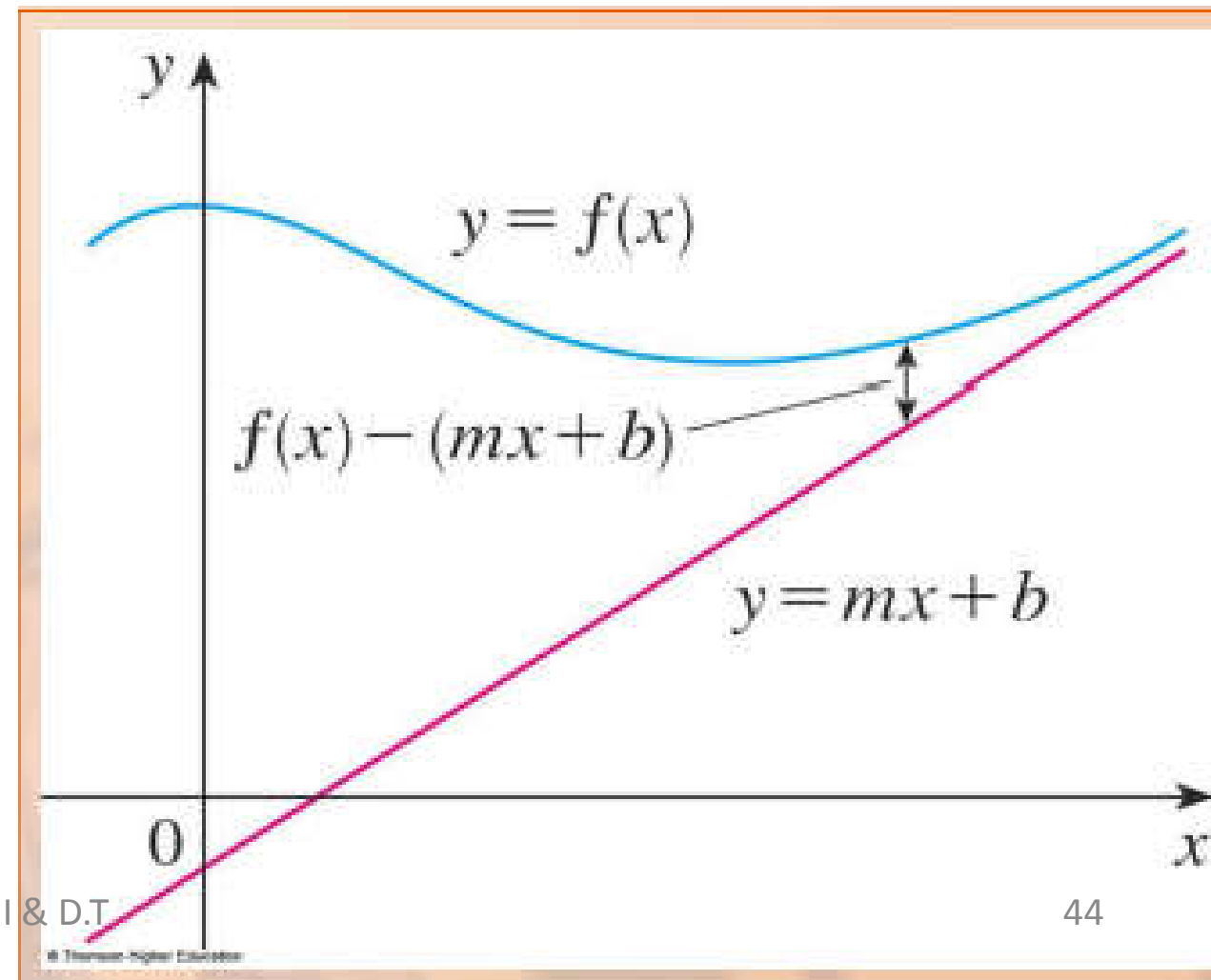


Oblique asymptote: Some curves have asymptotes that are oblique — that is, neither horizontal nor vertical

- For rational functions, oblique asymptotes occur when the degree of the numerator is one more than the degree of the denominator. In such a case, the equation of the oblique asymptote can be found by long division

- If $\lim_{x \rightarrow \pm\infty} [f(x) - (mx + b)] = 0$, then the line $y = mx + b$ is called **Oblique asymptote**.

This is because the vertical distance between the curve $y = f(x)$ and the line $y = mx + b \rightarrow 0$.



Example 9. Sketch the graph of: $f(x) = \frac{x^3}{x^2 + 1}$

Solution :

Step 1. The domain is: $\mathbb{R} = (-\infty, \infty)$

Step 2. As $f(-x) = -f(x)$, f is odd and its graph is symmetric about the origin

Step 3. The x - and y -intercepts are both 0.

Step 4. Since $x^2 + 1$ is never 0, there is **no vertical asymptote**.

Since $f(x) \rightarrow \infty$ as $x \rightarrow \infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, there is **no horizontal asymptote**

However, long division gives $f(x) = \frac{x^3}{x^2 + 1} = x - \frac{x}{x^2 + 1}$,

$$f(x) - x = -\frac{x}{x^2 + 1} =$$

$$= -\frac{1/x}{1 + 1/x^2} \rightarrow 0 \text{ as } x \rightarrow \pm\infty$$

So, the line $y = x$ is a oblique asymptote

Step 5. (derivatives). Find $f'(x)$ and $f''(x)$

$$f'(x) = \frac{3x^2(x^2 + 2) - x^3 \cdot 2x}{(x^2 + 1)^2} = \frac{x^2(x^2 + 3)}{(x^2 + 1)^2}$$

Since $f'(x) > 0$ for all x (except 0), f is increasing on $(-\infty, \infty)$

Although $f'(0) = 0$, f' does not change sign at 0

So, there is no local maximum or minimum

$$f''(x) = \frac{(4x^2 + 6x)(x^2 + 1)^2 - (x^4 + 3x^2) \cdot 2(x^2 + 1)2x}{(x^2 + 1)^4} = \frac{2x(3 - x^2)}{(x^2 + 1)^3}$$

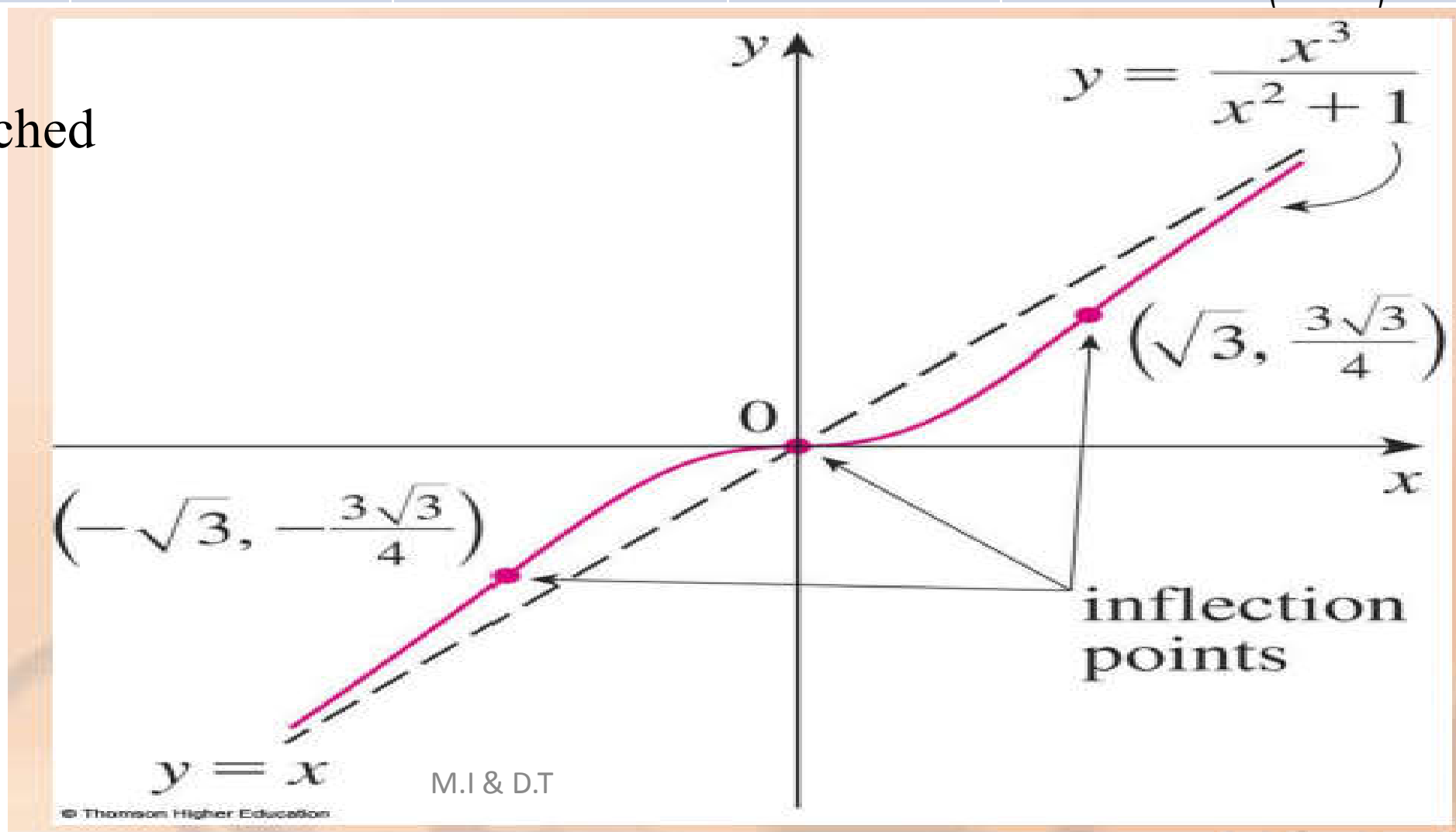
Since $f''(x) = 0$ when $x = 0$ or $x = \pm\sqrt{3}$, we set up the following chart

The points of inflection are: $(-\sqrt{3}, -\frac{3\sqrt{3}}{4})$, $(0, 0)$ and $(\sqrt{3}, \frac{3\sqrt{3}}{4})$

interval	x	$(3-x^2)$	$(x^2+1)^2$	$f''(x)$	f
$x < -\sqrt{3}$	—	—	+	+	CU on $(-\infty, -\sqrt{3})$
$-\sqrt{3} < x < 0$	—	+	+	—	CD on $(\sqrt{3}, 0)$
$0 < x < \sqrt{3}$	+	+	+	+	CU on $(0, \sqrt{3})$
$x > \sqrt{3}$	+	—	+	—	CD on $(\sqrt{3}, \infty)$

Step 6.

The graph of f is sketched



Activity 3. sketch the graph

$$\text{a) } f(x) = \frac{\cos x}{2 + \sin x}$$

$$\text{b) } f(x) = \frac{x^2}{\sqrt{x+1}}$$

$$\text{c) } f(x) = \frac{x^2 - 4}{x^2 - 9}$$

5.5. L'hôpital's Rule

Recall that a limit of the form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ [1]
where both $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$, then this limit may or may not exist
and is called an **indeterminate form of type 0/0**

Some examples encountered earlier in the previous session are:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

The first limit was obtained algebraically by factoring the numerator and canceling the common factor of $x - 1$, and the second two limits were obtained using geometric methods.

However, there are many indeterminate forms for which neither algebraic nor geometric methods will produce the limit.

Theorem: Suppose f and g are differentiable and $g(x) \neq 0$, on an open interval I that contain a (except possibly at a).

Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $0/0$ or ∞/∞)

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

If the limit on the right side exist or infinite (∞ **or** $-\infty$)

Applying L'Hôpital's Rule

Step 1. Check that the limit of $f(x)/g(x)$ is an indeterminate form of type $0/0$ or ∞/∞ .

Step 2. Differentiate f and g separately.

Step 3. Find the limit of $f'(x)/g'(x)$. If this limit is finite, ∞ *or* $-\infty$, then it is equal to the limit of $f(x)/g(x)$

Example 10. Find the limit $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$, using L'Hôpital's rule, and **check the result by factoring**

Solution: The numerator and denominator have a limit of 0, so the limit is an indeterminate form of type $0/0$.

Applying L'Hôpital's rule yields:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{\frac{d}{dx} [x^2 - 4]}{\frac{d}{dx} [x - 2]} = \lim_{x \rightarrow 2} \frac{2x}{1} = 4$$

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Activity 4.

In each part confirm that the limit is an indeterminate form of type 0/0, and evaluate it using **L'Hôpital's rule**

$$\text{a) } \lim_{x \rightarrow \infty} \frac{e^x + x^2}{e^x + 4x} \quad \text{b) } \lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} \quad \text{c) } \lim_{x \rightarrow \infty} \frac{x^{-4/3}}{\sin(1/x)} \quad \text{d) } \lim_{x \rightarrow 0^-} \frac{\tan x}{x^2}$$

$$\text{g) } \lim_{x \rightarrow 0} \frac{x + 6}{x + 2}$$

REMARK:

Applying L'Hôpital's rule to limits that are not indeterminate forms can produce **incorrect** results. For example, the computation of **g)**

Example 11. In each part confirm that the limit is an indeterminate form of type ∞ / ∞ and apply L'Hôpital's rule

a) $\lim_{x \rightarrow \infty} \frac{x}{e^x}$

b) $\lim_{x \rightarrow 0^+} \frac{\ln x}{\csc x}$

Solution:

a). The numerator and denominator both have a limit of ∞ , so we have an indeterminate form of type ∞ / ∞ .

Applying L'Hôpital's rule yields: $\lim_{x \rightarrow \infty} \frac{x}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$

b). The numerator has a limit of $-\infty$ and the denominator has a limit of ∞ , so we have an indeterminate form of type ∞ / ∞ . Applying L'Hôpital's rule yields:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\ln x}{\csc x} &= \lim_{x \rightarrow 0^+} \frac{1/x}{-\csc x \cot x} = \lim_{x \rightarrow 0^+} \left(-\frac{\sin x}{x} \tan x \right) \\ &= -\lim_{x \rightarrow 0^+} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0^+} \tan x = -(1)(0) = 0 \end{aligned}$$

Thus far we have discussed indeterminate forms of type $0/0$ and ∞ / ∞ . However, these are not the only possibilities; in general, the limit of an expression that has one of the forms

$$\frac{f(x)}{g(x)}, \quad f(x) \cdot g(x), \quad f(x)^{g(x)}, \quad f(x) + g(x), \quad f(x) - g(x)$$

is called an *indeterminate form* if the limits of $f(x)$ and $g(x)$ individually exert conflicting influences on the limit of the entire expression.

this can sometimes be evaluated by rewriting the product as a ratio, and then applying L'Hôpital's rule for indeterminate forms of type $0/0$ or ∞ / ∞ .

Example 12. Evaluate

a) $\lim_{x \rightarrow 0^+} x \ln x$

b) $\lim_{x \rightarrow \pi/4} (1 - \tan x) \sec 2x$

Solution:

a) The factor x has a limit of 0 and the factor $\ln x$ has a limit of $-\infty$, so the stated problem is an indeterminate form of type $0 \cdot \infty$. There are two possible approaches:

we can rewrite the limit as: $\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x}$ or $\lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}$

the first being an indeterminate form of type ∞ / ∞ and the second an indeterminate form of type $0/0$. However, the first form is the preferred initial choice because the derivative of $1/x$ is less complicated than the derivative of $1/\ln x$.

That choice yields:

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} &= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \\ &= \lim_{x \rightarrow 0^+} (-x) = 0 \end{aligned}$$

b) The stated problem is an indeterminate form of type $0 \cdot \infty$. We will

convert it to an indeterminate form of type $0/0$:

$$\begin{aligned} \lim_{x \rightarrow \pi/4} (1 - \tan x) \sec 2x &= \lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{1/\sec 2x} = \lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{\cos 2x} \\ &= \lim_{x \rightarrow \pi/4} \frac{-\sec^2 x}{-2 \sin 2x} \\ &= \frac{-2}{-2} = 1 \end{aligned}$$

Example 13. Evaluate $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$

Solution:

Both terms have a limit of $+\infty$, so the stated problem is an indeterminate form of type $\infty - \infty$. Combining the two terms yields:

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \sin x}$$

which is an indeterminate form of type $0/0$. Applying L'Hôpital's rule

$$\begin{aligned} \text{twice yields } \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x \sin x} &= \lim_{x \rightarrow 0^+} \frac{\cos x - 1}{\sin x + x \cos x} \\ &= \lim_{x \rightarrow 0^+} \frac{-\sin x}{\cos x + \cos x - x \sin x} \\ &= \lim_{x \rightarrow 0^+} \frac{-\sin x}{2 \cos x - x \sin x} = \frac{0}{2} = 0 \end{aligned}$$

Limits of the form $f(x)^{g(x)}$

Indeterminate forms of types 00 , 0 , and 1 can sometimes be evaluated by first introducing a dependent variable.

$$y = f(x)^{g(x)}$$

and then computing the limit of $\ln y$. Since

$$\ln y = \ln[f(x)^{g(x)}] = g(x) \cdot \ln f(x)$$

Example 14. Find $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$

Solution: As discussed above, we begin by introducing a dependent variable

$y = (1 + \sin x)^{1/x}$ and taking the natural logarithm of both sides

$$\ln y = \ln[(1 + \sin x)^{1/x}] = \frac{1}{x} \cdot \ln(1 + \sin x) = \frac{\ln(1 + \sin x)}{x}$$

Thus,
$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$$

which is an indeterminate form of type $0/0$, so by L'Hôpital's rule

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x} = \lim_{x \rightarrow 0} \frac{\cos x / (1 + \sin x)}{1} = 1$$

Since we have shown that $\ln y \rightarrow 1$ as $x \rightarrow 0$, the continuity of the exponential function implies that $e^{\ln y} \rightarrow e^1$ as $x \rightarrow 0$, and this implies that $y \rightarrow e$ as $x \rightarrow 0$.

Thus,
$$\lim_{x \rightarrow 0} (1 + \sin x)^{1/x} = e$$

Activity 5.

a) $\lim_{x \rightarrow \infty} (1 - 3/x)^x$ **b)** $\lim_{x \rightarrow 0} (1 - 2x)^{-3/x}$ **c)** $\lim_{x \rightarrow 0} (e^x + x)^{1/x}$

d) $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$ M.I. & D.T

THE END